ADVANCED NOTE-TAKING SYSTEM

7 Corrected Prompts — All 20 Mistakes Fixed

Block Editor • 35+ Block Types • Inline Databases • AI Coach • 25 Unique Features

---

Summary of All 20 Fixes Applied

Fix #	Affected	What Was Fixed
Fix 1	All Prompts	Corrected tech stack: React + Vite + Drizzle + Express (not Next.js/Prisma)
Fix 2	Prompt 5	Added note about existing 'notes' table to avoid conflicts
Fix 3	All Prompts	Changed src/components/notes/ → client/src/components/notes/
Fix 4	Prompt 5	Changed Prisma schema syntax → Drizzle pgTable() syntax
Fix 5	Prompt 5	Added explicit 'npx drizzle-kit push' command
Fix 6	Prompt 5	Added uuid, jsonb, date imports to shared/schema.ts
Fix 7	Prompt 5	Standardized PKs to uuid().defaultRandom().primaryKey() with import
Fix 8	Prompt 5	Changed user FK columns from uuid → text (matches Replit Auth)
Fix 9	All Prompts	Replaced workspace_id with userId (no workspace system exists)
Fix 10	Prompts 5-6	Added API route instructions for server/routes.ts
Fix 11	Prompt 1	Added route registration in client/src/App.tsx with wouter
Fix 12	Prompts 5-6	Added IStorage interface pattern instructions
Fix 13	Prompt 5	Fixed jsonb defaults to use .$defaultFn(() => ({}))
Fix 14	Prompt 3	Added KaTeX package install instruction
Fix 15	Prompt 4	Added PDF export approach (window.print() or jspdf)
Fix 16	Prompt 3	Changed eval() → sandboxed iframe with srcdoc
Fix 17	Prompt 3	Added HTTPS/mic permission note for voice recording
Fix 18	Prompts 3,6	Fixed content recycler query for NULL last_visited_at
Fix 19	Prompt 5	Added npx drizzle-kit push as required step in boundaries
Fix 20	Prompt 1	Changed hardcoded 'My Workspace' → user's name from auth

PROMPT 1 OF 7

ALL UI STRUCTURE: Sidebar + Page + Settings + Editor + Slash + Toolbar + Database + Comments + Search + Focus + Split + ContentMap

⚡ **REPLIT: LITE MODE (lightning bolt) | App Testing: OFF | Code Optimization: OFF**

> ✅ **FIX APPLIED:** Fix 1 — Tech stack corrected to React + Vite + Drizzle + Express
> ✅ **FIX APPLIED:** Fix 3 — All file paths corrected to client/src/components/notes/
> ✅ **FIX APPLIED:** Fix 9 — workspace_id removed; uses userId from auth
> ✅ **FIX APPLIED:** Fix 11 — Route registration added for App.tsx
> ✅ **FIX APPLIED:** Fix 20 — 'My Workspace' replaced with user's display name from auth

```
CONTEXT:
App: AI-driven productivity platform with auth, teams, tasks
Tech: React, TypeScript, Vite, Tailwind CSS, Shadcn/UI, PostgreSQL, Drizzle ORM,
Express.js backend in server/routes.ts
Building: Advanced note-taking system with block editor, 35+ block types, databases,
AI coaching, focus mode, voice notes, sketching, and 25 unique features beyond
competitors
Feature: ALL 12 UI component shells — structure only, no interactivity
Current state: App has auth. No notes system exists. No workspace system exists yet.
Layer: Layer 1 — Structure only (12 components)

Files to create:
client/src/components/notes/Sidebar.tsx
client/src/components/notes/PageView.tsx
client/src/components/notes/PageSettingsPanel.tsx
client/src/components/notes/BlockEditor.tsx
client/src/components/notes/SlashCommandGrid.tsx
client/src/components/notes/BlockToolbar.tsx
client/src/components/notes/DatabaseView.tsx
client/src/components/notes/CollaborationPanel.tsx
client/src/components/notes/SearchCommandPalette.tsx
client/src/components/notes/FocusMode.tsx
client/src/components/notes/SplitView.tsx
client/src/components/notes/ContentMap.tsx

Also: Register a route in client/src/App.tsx: <Route path="/dashboard/notes"
component={NotesPage} /> using wouter's Route component.

TASK:
Build all 12 UI shells with hardcoded data. Nothing interactive. No click handlers,
no state changes, no API calls.
```

=== COMPONENT 1: SIDEBAR (Sidebar.tsx) ===

Left panel (w-[260px] h-full bg-gray-50 border-r border-gray-200 flex flex-col).

Top: Display the user's name from auth (font-semibold text-sm px-4 py-3) + collapse icon. (Use the authenticated user's display name instead of a hardcoded workspace name — workspaces are not built yet.)

Tab bar (flex border-b px-2 gap-1): 4 tabs "Pages"/"Recent"/"Favorites"/"Tags".

Active tab: bg-white rounded-t-md shadow-sm text-blue-700 text-xs font-medium px-3 py-1.5.

Inactive: text-gray-500 text-xs px-3 py-1.5 hover:bg-gray-100 rounded-t-md.

"Pages" tab content (flex-1 overflow-y-auto px-2 py-2):

Page tree with 6 hardcoded pages with indentation:

"Getting Started" (icon 📖, no children)

"Project Notes" (icon 📁, expanded, 2 children indented pl-6: "Sprint 1" "Sprint 2")

"Meeting Notes" (icon 📋, collapsed with expand arrow)

"Ideas" (icon 💡), "Research" (icon 🔍), "Archive" (icon 📦 text-gray-400)

Each row: h-8 px-2 rounded-md hover:bg-gray-100 cursor-pointer flex items-center gap-2 text-sm.

Expand arrow (ChevronRight w-3 h-3) + icon (text-base) + name (truncate flex-1).

Selected page: bg-blue-50 text-blue-700.

Quick Capture inbox section (border-t px-3 py-2): "Inbox" header (text-xs text-gray-400 uppercase) + 2 hardcoded captured items (text-xs text-gray-600 truncate py-1).

Bottom: "+ New Page" button (w-full h-9 flex items-center gap-2 px-4 text-sm text-gray-500 hover:bg-gray-100 rounded-md) with Plus icon.

"Recent" tab: 5 page names with time "2h ago" "Yesterday".

"Favorites" tab: 3 starred pages with Star icon.

"Tags" tab: 6 tag pills "Work" "Personal" "Ideas" "Research" "Team" "Archive" (bg-gray-100 text-xs rounded-full px-2 py-1).

Show only "Pages" tab content visible.

=== COMPONENT 2: PAGE VIEW (PageView.tsx) ===

Main content area (flex-1 max-w-[900px] mx-auto px-16 py-8).

Cover image area (h-[200px] w-full bg-gradient-to-r from-blue-400 to-purple-500 rounded-b-lg relative). "Change cover" link (absolute bottom-3 right-3 text-white/70 text-xs).

Icon (text-5xl -mt-8 ml-2 relative z-10 cursor-pointer) showing 📖.

Breadcrumbs (flex items-center gap-1 text-xs text-gray-400 mt-3): "Notes" > "Project Notes" > "Sprint 1".

Status badge inline (bg-yellow-100 text-yellow-700 text-xs px-2 py-0.5 rounded-full ml-2): "In Progress".

Reading stats (text-xs text-gray-400 ml-auto): "342 words • 2 min read • Last edited 3h ago".

Title (text-4xl font-bold text-gray-900 mt-2 outline-none placeholder "Untitled" w-full border-0).

Content area below (mt-6 min-h-[400px]): Placeholder blocks:

Paragraph text block (text-base text-gray-700 py-1).

Heading 2 "Key Decisions" (text-2xl font-semibold py-2).

Bullet list: 3 items (pl-6).

Callout block (bg-blue-50 border-l-4 border-blue-400 p-4 rounded-r-lg flex gap-3): 💡 icon + text.

Code block (bg-gray-900 text-green-400 rounded-lg p-4 font-mono text-sm): 3 lines of sample code with language badge "javascript" top-right.

Each block has invisible grip handle area on left (opacity-0 group-hover:opacity-100 w-6).

=== COMPONENT 3: PAGE SETTINGS PANEL (PageSettingsPanel.tsx) ===

Right sidebar (fixed right-0 top-0 h-full w-[340px] bg-white shadow-xl border-l z-40 overflow-y-auto).

Header: "Page Settings" (font-semibold text-lg) + X close (border-b px-5 py-4).

Scrollable sections (px-5 py-4 space-y-6):

"Appearance" section (text-xs font-semibold text-gray-400 uppercase mb-3):

Font selector: 3 cards side-by-side (grid grid-cols-3 gap-2): "Default"(active, blue border) / "Serif" / "Mono" — each showing Aa preview.

"Small text" toggle (flex justify-between py-2). "Full width" toggle.

"Page Status" section: 5 status pills (flex flex-wrap gap-2): "Draft"(gray)/"In Progress"(yellow)/"Review"(blue)/"Final"(green)/"Archived"(gray-400).

"Page Info" section: Icon picker button, Cover image button, Created "Mar 1, 2026", Last edited "3h ago", Word count "342", Reading time "2 min".

"AI Insights" section: "Generate Insights" button (border text-sm) + placeholder "Insights will appear here".

"Actions" section: Rows (each h-10 flex items-center hover:bg-gray-50 rounded px-2 text-sm cursor-pointer): Copy link, Duplicate page, Move to..., Lock page, Suggest edits, Use with AI, Translate, History, Share, Export, Import.

"Danger Zone" section: "Move to Trash" (text-red-500 text-sm).

=== COMPONENT 4: BLOCK EDITOR (BlockEditor.tsx) ===

Container for rendering blocks (w-full min-h-[400px]).

10 hardcoded sample blocks demonstrating different types:

Block 1: TEXT — regular paragraph (text-base text-gray-700 py-1 px-1).

Block 2: HEADING 2 — "Project Overview" (text-2xl font-semibold py-2).

Block 3: BULLET LIST — 3 items (pl-6, each with • marker).

Block 4: TO-DO — 2 checkboxes (first checked with line-through).

Block 5: TOGGLE — "Click to expand" with ChevronRight + hidden content.

Block 6: QUOTE — left border (border-l-3 border-gray-300 pl-4 italic text-gray-600).

Block 7: CALLOUT — (bg-amber-50 border-l-4 border-amber-400 p-3 rounded-r-lg flex gap-3) ⚠️ icon + warning text.

Block 8: CODE — (bg-gray-900 text-green-400 rounded-lg p-4 font-mono text-sm overflow-x-auto) with "python" badge.

Block 9: DIVIDER — (border-t border-gray-200 my-4).

Block 10: TABLE — 3x3 grid (border border-gray-200, each cell px-3 py-2 text-sm border).

Each block wrapped in container (group relative py-0.5): Left grip handle (absolute -left-8 top-1/2 -translate-y-1/2 opacity-0 group-hover:opacity-100 w-6 h-6 flex items-center justify-center text-gray-300 cursor-grab): GripVertical icon. Block type icon (absolute -left-14 opacity-0 group-hover:opacity-40 text-xs text-gray-400).

Empty block row at bottom: just cursor line blinking.

=== COMPONENT 5: SLASH COMMAND GRID (SlashCommandGrid.tsx) ===

Popup positioned below cursor (absolute z-50 w-[480px] bg-white rounded-xl shadow-2xl border p-4).

Search input at top (h-9 w-full border-0 outline-none text-sm placeholder "Search blocks...").

Category headers (text-xs font-semibold text-gray-400 uppercase mt-3 mb-2).

"BASIC" category: Card grid (grid grid-cols-3 gap-2) of block type cards. Each card (p-3 rounded-lg hover:bg-blue-50 cursor-pointer border border-transparent hover:border-blue-200 transition): Icon (text-xl mb-1) + name (text-sm font-medium) + desc (text-xs text-gray-400).

Cards: 📝 Text, 🆗 Heading 1, 🆘 Heading 2, 🆙 Heading 3, • Bullet List, 1. Numbered, ☑ To-Do, ▸ Toggle, " Quote.

"MEDIA" category: 🖼️ Image, 🎥 Video, 🎤 Voice Note, ✍️ Sketch, 📎 File, 🔗 Bookmark, 🌐 Embed.

"ADVANCED" category: 📊 Table, 🗃️ Database, 💻 Code, ✖ Equation, 💡 Callout, — Divider, 🔄 Synced Block.

"UNIQUE" category (with ✨ sparkle accent): 🎯 Progress Bar, 📚 Flashcard, 📅 Timeline, 🗳️ Poll, 📚 Reading List, ▸ Code Runner.

Show popup visible for testing.

=== COMPONENT 6: BLOCK TOOLBAR (BlockToolbar.tsx) ===

Floating bar that appears above text selection (absolute z-50).

Bar (bg-gray-900 rounded-lg shadow-xl px-2 py-1 flex items-center gap-0.5).

Format buttons (each w-7 h-7 rounded flex items-center justify-center text-white/80 hover:bg-white/20 text-sm): B(bold), I(italic), U(underline), S(strikethrough), <>(code), 🔗(link), A with color dot(text color picker), highlight icon(background color), @(mention), 💬(comment on selection), ✨(AI improve).

Separator line (w-px h-4 bg-white/20) between format group and action group.

Action buttons: "Turn into" dropdown trigger, "Copy" icon.

Show bar visible for testing at a fixed position.

=== COMPONENT 7: DATABASE VIEW (DatabaseView.tsx) ===

Inline block (bg-white rounded-lg border border-gray-200 overflow-hidden).

Header (px-4 py-2 border-b flex items-center justify-between): Database title "Task Tracker" (font-medium text-sm) + view tabs (flex gap-1):

"Table"(active)/"Board"/"Calendar"/"Timeline"/"Gallery"/"List" (each text-xs px-2 py-1 rounded).

Toolbar (px-4 py-1.5 border-b flex gap-2 text-xs text-gray-500): "Filter" + "Sort" + "Group" + "Properties" + "New view" + Search icon.

TABLE VIEW content: Header row (bg-gray-50 text-xs font-medium text-gray-500): Name / Status / Priority / Assignee / Due Date. 5 data rows (h-10 border-b text-sm hover:bg-gray-50). Status as colored badge. Priority as icon. Assignee as avatar. Date as text.

BOARD VIEW placeholder (flex gap-4 p-4): 3 columns "To Do"/"In Progress"/"Done" with 2-3 cards each.

"+ New" button at bottom (text-sm text-gray-400 px-4 py-2).

Show Table view active.

=== COMPONENT 8: COLLABORATION PANEL (CollaborationPanel.tsx) ===

Right panel (w-[320px] bg-white border-l h-full overflow-y-auto).

Tabs: "Comments"(active) / "Activity" (text-sm).

"Comments" tab: 3 hardcoded comment threads. Each: avatar (w-8 h-8 rounded-full bg-gray-200) + name (font-medium text-sm) + time (text-xs text-gray-400) + comment text (text-sm mt-1) + "Reply" link (text-xs text-blue-600) + "Resolve" link (text-xs text-gray-400).

Block reactions section: "Reactions on this page" header (text-xs text-gray-400). 2 reaction rows: block text snippet + emoji reactions (👍 3, ❤️ 1).

Comment input at bottom (border-t p-3): avatar + input (rounded-lg border text-sm placeholder "Write a comment...").

"Activity" tab: 5 entries "Alice edited Block 3" "Bob added comment" etc with timestamps.

=== COMPONENT 9: SEARCH COMMAND PALETTE (SearchCommandPalette.tsx) ===

Overlay (fixed inset-0 bg-black/40 z-50 flex items-start justify-center pt-[15vh]).

Card (w-[600px] bg-white rounded-xl shadow-2xl overflow-hidden).

Input (h-12 w-full border-0 px-5 text-base placeholder "Search pages, blocks, or type a command..." outline-none).

Divider.

Quick actions section (px-4 py-2): "ACTIONS" header (text-xs text-gray-400). Action rows: "+ New Page" / "Quick Capture" / "Toggle Focus Mode" / "Open Split View" / "Open Content Map" (each h-9 px-3 rounded-md hover:bg-gray-50 flex items-center gap-3 text-sm cursor-pointer).

Recent section: "RECENT" header + 4 page rows with icon + title + time.

Search results placeholder: "Start typing to search..." (text-sm text-gray-400 text-center py-6).

Keyboard hint: "Ctrl+K" (text-xs text-gray-300 px-4 py-2 text-right).

Show visible for testing.

=== COMPONENT 10: FOCUS MODE (FocusMode.tsx) ===

Full-screen overlay (fixed inset-0 bg-gray-950 z-50 flex flex-col).

Top bar (flex justify-between items-center px-8 py-4):

Left: "Focus Mode" (text-white/60 text-sm) + word target "Goal: 500 words".

Center: Animated progress ring (w-12 h-12 SVG circle stroke-width-3 stroke-blue-400) with "342" inside (text-white text-xs).

Right: Ambient sound selector (3 icons: Rain/Forest/Silence, text-white/40) + timer "25:00" + X exit (text-white/40).

Editor area (max-w-[700px] mx-auto flex-1 py-12):

Title "Sprint 1 Planning" (text-3xl font-bold text-white outline-none placeholder "Start writing...").

Content (text-lg text-white/80 leading-relaxed mt-6): 2 paragraphs of sample text. Current block slightly brighter (text-white), other blocks dimmed (text-white/40).

Bottom (text-center py-4): Word count "342 / 500" (text-white/30 text-sm) + "Press Esc to exit" (text-white/20 text-xs).

Show visible for testing.

=== COMPONENT 11: SPLIT VIEW (SplitView.tsx) ===

Full layout (flex h-full).

Left pane (flex-1 border-r border-gray-200 overflow-y-auto): PageView header with "Sprint 1" + content blocks.

Divider handle (w-1 bg-gray-200 hover:bg-blue-400 cursor-col-resize relative flex items-center justify-center): Grip dots (w-3 h-8 flex flex-col gap-0.5 items-center justify-center): 6 small dots (w-1 h-1 rounded-full bg-gray-400).

Right pane (flex-1 overflow-y-auto): PageView header with "Research Notes" + different content blocks.

Tab bar on each pane top (h-8 bg-gray-50 border-b flex items-center px-3 gap-2): Page title (text-xs font-medium) + X close (text-gray-400 w-4 h-4).

=== COMPONENT 12: CONTENT MAP (ContentMap.tsx) ===

Full overlay (fixed inset-0 bg-white z-50 flex flex-col).

Header (h-12 border-b px-6 flex items-center justify-between): "Content Map" (font-semibold) + zoom controls (+/- buttons) + X close.

Canvas area (flex-1 overflow-auto p-8 bg-gray-50):

Mind-map layout with 7 hardcoded page nodes:

Center node "My Notes" (bg-blue-500 text-white rounded-xl px-4 py-2 shadow-lg font-medium).

Connected nodes radiating out: "Project Notes" "Meeting Notes" "Ideas" "Research" "Getting Started" "Archive".

Each node (bg-white rounded-lg shadow-md px-3 py-2 text-sm border border-gray-200 cursor-pointer hover:shadow-lg).

Connection lines (SVG paths, stroke-gray-300 stroke-width-2) between linked pages.

Page icon displayed on each node.

Show visible for testing.

EMPTY STATES:

- No pages: "Create your first page to get started"
- Favorites empty: "Star pages to add them here"
- Tags empty: "AI will auto-tag your pages, or add tags manually"

BOUNDARIES — CRITICAL:

- Do not add any click handlers or state changes
- Do not add any API calls or database connections
- Do not modify any existing files (except adding the route in App.tsx)
- Build only and exactly what is described above — all 12 components

PROMPT 2 OF 7

LOGIC: Sidebar + Page Controls + Editor Core Blocks

🔵 **REPLIT: AUTONOMOUS → ECONOMY | App Testing: OFF | Code Optimization: OFF**

> ✅ **FIX APPLIED:** Fix 1 — Tech stack corrected
> ✅ **FIX APPLIED:** Fix 3 — File paths corrected to client/src/components/notes/
> ✅ **FIX APPLIED:** Fix 9 — No workspace_id references

```
CONTEXT:
App: AI-driven productivity platform with auth, teams, tasks
Tech: React, TypeScript, Vite, Tailwind CSS, Shadcn/UI, PostgreSQL, Drizzle ORM,
Express.js backend in server/routes.ts
Building: Advanced note-taking system with block editor, 35+ block types, databases,
AI coaching, focus mode, voice notes, sketching, and 25 unique features beyond
competitors
Feature: All sidebar interactions + page controls + core block editor with basic
block types, slash commands, drag-drop, rich text, markdown, keyboard shortcuts
Current state: Prompt 1 done. All 12 UI shells exist.
Layer: Layer 2+3 — All interactions (no database)
Files to change: client/src/components/notes/Sidebar.tsx,
client/src/components/notes/PageView.tsx,
client/src/components/notes/PageSettingsPanel.tsx,
client/src/components/notes/BlockEditor.tsx,
client/src/components/notes/SlashCommandGrid.tsx,
client/src/components/notes/BlockToolbar.tsx

=== PART A: SIDEBAR + PAGE CONTROLS ===
1. SIDEBAR TABS: Clicking tab switches content. Pages/Recent/Favorites/Tags. Active
tab styling.
2. PAGE TREE: Click page selects it and loads in PageView. Expand/collapse arrow
toggles children (200ms). Drag page rows to reorder (onDragStart/onDragOver/onDrop).
Right-click page shows context menu: Rename/Duplicate/Move/Delete/Add to
Favorites/Set Status.
3. NEW PAGE: "+ New Page" creates untitled page in state, selects it, focuses title.
4. FAVORITES: Star icon on hover toggles favorite. Favorites tab shows starred
pages.
5. TAGS: Click tag filters pages by that tag. "Add Tag" on page adds to tags state.
6. QUICK CAPTURE: Capture button opens mini input (overlay bottom-right). Type text
+ Enter saves to inbox array. Inbox items shown in sidebar.
7. RECENT: Track last 10 pages opened. Show in Recent tab with relative time.
8. SEARCH: Ctrl+K opens SearchCommandPalette. Escape closes. Type filters pages by
title. Actions execute: "New Page" creates, "Focus Mode" opens, "Split View" opens,
"Content Map" opens.
9. PAGE VIEW: Title contentEditable. Click to edit, blur saves. Breadcrumbs
auto-generated from page parent chain.
```

10. COVER: "Change cover" opens picker with 6 gradient presets + "Upload" option + "Remove" option. Selecting changes cover in state.

11. ICON: Click icon opens emoji picker grid (6x8 common emojis). Selecting changes icon.

12. STATUS: Click badge opens dropdown: Draft(gray)/In Progress(yellow)/Review(blue)/Final(green)/Archived(gray-400). Updates in sidebar tree too.

13. READING STATS: Calculate word count from blocks, reading time (words/200 rounded up), last edited timestamp. Display inline.

14. SETTINGS PANEL: Gear icon or three-dot menu opens panel (slide 200ms). X/outside closes.

15. FONT: Clicking Default/Serif/Mono card changes page font-family in state.

16. SMALL TEXT: Toggle reduces base font size from 16px to 14px.

17. FULL WIDTH: Toggle changes max-width from 900px to 100%.

18. LOCK: Toggle disables all editing on page content. Shows lock icon in header.

19. ACTIONS: Copy Link copies URL to clipboard + toast. Duplicate creates copy. Move opens page picker. Delete moves to trash array. Undo restores last action.

20. History: Opens version list (placeholder, populated in DB prompts).

21. Share: Opens share modal with link copy + permission dropdown.

22. Export: Downloads page as Markdown. Import: Accepts .md file, parses to blocks.

=== PART B: EDITOR CORE BLOCKS ===

23. BLOCK DATA MODEL (in React state): Array of block objects. Each: { id: uuid, type: string, content: string, properties: {}, children: [] (for nested/toggle), indent: 0-3 }

24. BASIC BLOCK TYPES (render based on type):

"text" — contentEditable div (text-base). Enter at end creates new text block below. Backspace on empty deletes block.

"heading1" — text-3xl font-bold. "heading2" — text-2xl font-semibold. "heading3" — text-xl font-medium.

"bullet_list" — • marker + text (pl-6). Enter creates next bullet. Empty + Enter converts to text.

"numbered_list" — Auto-incrementing number. Same enter behavior.

"todo" — Checkbox + text. Click checkbox toggles checked (line-through + text-gray-400). Enter creates next todo.

"toggle" — ChevronRight/Down icon + text. Click arrow expands/collapses children area (200ms). Children can be any block type.

"toggle_heading1/2/3" — Heading + toggle combined.

"quote" — border-l-3 border-gray-300 pl-4 text-gray-600 italic.

"callout" — bg-[color]-50 border-l-4 border-[color]-400 p-3 rounded-r-lg. Icon picker + color picker. Default: 💡 blue.

"divider" — border-t border-gray-200 my-3. Not editable.

"code" — bg-gray-900 text-green-400 rounded-lg p-4 font-mono text-sm. Language selector dropdown (50+ languages). Copy button top-right.

"table" — Rows x Cols grid. Tab moves between cells. Shift+Tab goes back. Enter in last cell adds row. Column resize handles. Header row toggle.

"image" — Click to upload or paste URL. Display with resize handles (drag corners). Caption text below. Alignment: left/center/full-width.

"video" — Paste URL (YouTube/Vimeo). Renders embedded player.

"file" — Upload attachment. Shows filename + size + download icon.

"bookmark" — Paste URL. Renders card with site title + description + favicon + thumbnail.

"embed" — Paste URL. Renders iframe. Resize handles.

25. COLUMN LAYOUT: Type /columns or drag block to side of another block. Creates 2-column layout (flex gap-4). Each column is a drop zone for blocks. Resize handle between columns.

26. SLASH COMMANDS:

Typing "/" at start of empty block opens SlashCommandGrid below cursor. Search input filters cards by name. Arrow keys navigate. Enter selects. Selecting converts current block to that type. Typing more characters after "/" filters: "/h1" shows Heading 1, "/todo" shows To-Do, etc. Escape or click outside closes grid.

27. BLOCK TOOLBAR:

Select text in any block → floating toolbar appears above selection. B toggles bold (Ctrl+B). I toggles italic (Ctrl+I). U toggles underline (Ctrl+U). S toggles strikethrough (Ctrl+Shift+S). <> toggles inline code (Ctrl+E). Link opens URL input (Ctrl+K). Text color: Click A opens color grid (10 colors + default). Background: Same grid for highlight. "Turn into" dropdown: Convert selected block to any other type while keeping content.

28. DRAG-DROP:

Hover block → grip handle appears on left. Drag grip to reorder blocks. Blue insertion line shows drop position. Drop reorders in state array. Drag to column edge creates/moves to column layout.

29. MARKDOWN SHORTCUTS (on typing):

"# " at line start → heading1. "## " → heading2. "### " → heading3.

"- " or "* " → bullet_list. "1. " → numbered_list. "[] " or "[ ] " → todo.

"> " → quote. Triple backticks → code block. "---" → divider.

"**text**" → bold. "*text*" → italic. "~~text~~" → strikethrough. Backtick-wrapped text → inline code.

30. KEYBOARD SHORTCUTS:

Ctrl+Z undo, Ctrl+Y redo (maintain 50-step history stack). Tab indents block (max 3 levels). Shift+Tab outdents. Ctrl+Shift+0 → text, Ctrl+Shift+1 → h1, Ctrl+Shift+2 → h2, Ctrl+Shift+3 → h3. Ctrl+Shift+4 → todo, Ctrl+Shift+5 → bullet, Ctrl+Shift+6 → numbered. Ctrl+D duplicate block. Ctrl+Shift+Delete delete block. Ctrl+Up/Down move block. Ctrl+/ toggle block menu.

31. BLOCK MENU: Click grip handle (or right-click block or type Ctrl+/) opens block context menu: "Delete" / "Duplicate" / "Turn into..." / "Move to..." / "Copy link to block" / "Color" / "Comment" / "Add reaction"

BOUNDARIES — CRITICAL:

- Do not add any database reads or writes
- All data in React state only (lost on refresh — expected)
- Do not change any visual design from Prompt 1
- Do not add any features not described above

PROMPT 3 OF 7

LOGIC: Advanced Blocks + Unique Blocks + AI + Focus + All Unique Features

🔵 **REPLIT: AUTONOMOUS → ECONOMY | App Testing: OFF | Code Optimization: OFF**

> ✅ **FIX APPLIED:** Fix 1 — Tech stack corrected
> ✅ **FIX APPLIED:** Fix 3 — File paths corrected
> ✅ **FIX APPLIED:** Fix 14 — KaTeX install instruction added
> ✅ **FIX APPLIED:** Fix 16 — Code Runner uses sandboxed iframe instead of eval()
> ✅ **FIX APPLIED:** Fix 17 — Voice note HTTPS/mic permission note added
> ✅ **FIX APPLIED:** Fix 18 — Content recycler handles NULL last_visited_at

```
CONTEXT:
App: AI-driven productivity platform with auth, teams, tasks
Tech: React, TypeScript, Vite, Tailwind CSS, Shadcn/UI, PostgreSQL, Drizzle ORM, Express.js backend in server/routes.ts
Building: Advanced note-taking system with block editor, 35+ block types, databases, AI coaching, focus mode, voice notes, sketching, and 25 unique features beyond competitors
Feature: All special block types + 10 unique blocks + AI writing coach + smart links + focus mode + split view + presentation mode + snippet library + content recycler + daily journal + page insights + smart tags + page comparison + content map
Current state: Prompt 2 done. Sidebar, page controls, core editor work.
Layer: Layer 2+3
Files to change: client/src/components/notes/BlockEditor.tsx, client/src/components/notes/SlashCommandGrid.tsx, client/src/components/notes/PageView.tsx, client/src/components/notes/PageSettingsPanel.tsx, client/src/components/notes/FocusMode.tsx, client/src/components/notes/SplitView.tsx, client/src/components/notes/ContentMap.tsx, client/src/components/notes/Sidebar.tsx

PREREQUISITE: Install katex package first — add katex to dependencies.

=== PART A: REMAINING STANDARD BLOCKS ===
1. EQUATION BLOCK: Renders LaTeX/KaTeX math using the katex package. Click to edit raw LaTeX, blur renders formula. Inline equations with $formula$ syntax.
2. TABLE OF CONTENTS: Auto-generates from all headings on the page. Updates live as headings change. Click entry scrolls to that heading. Indented by heading level.
3. MENTION (inline): Type @ to trigger user mention popup. Type [[ to trigger page mention popup. Type @date to insert date picker. Mentions render as colored pill links.
4. SYNCED BLOCK: Create from block menu "Sync this block". Produces a synced copy that stays identical. Edit one, all copies update. Orange border indicator. "Unsync" option.
```

5. TEMPLATE BUTTON BLOCK: Click creates pre-defined block structure. Configure: button label + template blocks array. Example: "Add Meeting Notes" button creates heading + date + attendees list + action items.

6. LINK TO PAGE: Type /link or paste internal page URL. Renders as a card with page icon + title + preview text. Click navigates to page.

7. LINK PREVIEW: Paste external URL on its own line. Auto-fetches and renders rich preview card (title, description, image, favicon).

=== PART B: 10 UNIQUE BLOCKS (not in competitors) ===

8. VOICE NOTE BLOCK: Type /voice. Shows record button (red circle). Click starts recording (animated pulse, timer counting). Click stop saves audio. Shows waveform visualization + play/pause + duration. "Transcribe" button generates text transcription below (mock AI for now). Transcription is searchable. NOTE: Voice recording requires HTTPS and user microphone permission — this will work after deployment but may not work in Replit's dev preview.

9. SKETCH BLOCK: Type /sketch. Opens canvas (h-[300px] bg-white border rounded-lg). Tools: pen (3 sizes), eraser, color picker (8 colors), undo, redo, clear. Draw with mouse/touch. Save as SVG in block state.

10. FLASHCARD BLOCK: Type /flashcard. Card with front/back. Click "Flip" toggles (3D flip animation 300ms). Edit front (question) and back (answer). "Add to study deck" link. Multiple flashcards stack as a deck.

11. PROGRESS TRACKER BLOCK: Type /progress. Shows: label input "Sprint Progress" + current value slider (0-100) + color selector (green/blue/orange/red). Renders as rounded bar with percentage text + label.

12. READING LIST BLOCK: Type /readinglist. Add URL input. Shows card: favicon + title + domain + "Read"/"Unread" toggle badge. Stack multiple URLs as a list.

13. TIMELINE BLOCK: Type /timeline. Horizontal scrollable timeline. Each entry: date + title + description. "Add entry" button. Entries shown as dots on a horizontal line with content below.

14. POLL BLOCK: Type /poll. Question input + option inputs (2-6 options). "Add option" link. Shows results bar chart after voting. Click option to vote. Show counts.

15. CODE RUNNER BLOCK: Type /run. Code editor (like code block) + "Run" button below. Executing shows output in a result area below code. Support JavaScript execution using a sandboxed iframe with srcdoc (do NOT use eval() as it may be blocked by CSP). Display console.log output captured from the iframe.

16. BLOCK REACTIONS: Any block. Hover shows small "+" emoji button (bottom-right of block). Click opens emoji picker (12 common emojis). Selecting adds reaction. Reactions display below block as pills with count. Click existing reaction to add/remove your vote.

17. BLOCK BOOKMARKS: From block menu select "Bookmark this block". Block saved to user's bookmark collection (viewable from sidebar). Bookmarks reference the original block and page.

=== PART C: AI + FOCUS + ALL UNIQUE FEATURES ===

18. AI WRITING COACH: Toggle in settings "AI Coach". When ON: Right margin shows suggestion cards (bg-blue-50 rounded-lg p-2 text-xs max-w-[200px]): "This sentence could be clearer" / "Consider a stronger verb" / "Reading level: Grade 8". Cards appear contextually as user types (debounced 2s after stop typing). Mock AI: generate 2-3 random suggestions from a preset list based on text length.

19. SMART LINKS: As user types, detect words/phrases that match titles of other pages. Show subtle blue underline with tooltip "[Page Name] exists — click to link". Clicking inserts a page mention link. Check every 3 seconds (debounced).

20. FOCUS MODE: Triggered from command palette or sidebar button or Ctrl+Shift+F. Hides sidebar, covers everything with dark bg. Only shows title + content centered. Current block text-white, other blocks text-white/30. Word count progress ring animates. Exit with Escape or X. Ambient sound buttons play/stop (mock: just toggle icon state). Timer counts down from configurable time (default 25 min).

21. SPLIT VIEW: Triggered from command palette or Ctrl+\. Opens current page on left, page picker for right. Divider draggable to resize (min 30% each side). Each pane independently scrollable. X on pane tab closes split view.

22. PRESENTATION MODE: Triggered from page menu "Present". Builds slides from page: each H1 or H2 starts a new slide. Content between headings = slide body. Full-screen overlay. Arrow keys or click navigate slides. Slide counter "3/12" bottom-right. Escape exits.

23. SNIPPET LIBRARY: Accessible from sidebar "Snippets" or /snippet command. Save: Select blocks → block menu "Save as snippet" → name it. Insert: /snippet opens library, click inserts blocks at cursor. Store as array of block objects in state.

24. CONTENT RECYCLER: Surfaces pages not visited in 30+ days. Query: WHERE (last_visited_at < 30 days ago OR last_visited_at IS NULL) AND word_count > 50. Shows in sidebar as "Rediscover" section (collapsible, max 3 items). Each: page icon + title + "Last visited 45 days ago" (or "Never visited" if null). Click opens page. Dismiss removes from recycler.

25. DAILY JOURNAL: Sidebar button "Today's Journal". Creates/opens today's journal page with template: Date heading + Mood selector (5 emojis: 😄🙂😐😟😢) + "What's on your mind?" prompt + Gratitude section + Goals section. Streak counter: "7 day streak 🔥" in sidebar.

26. PAGE INSIGHTS: "Generate Insights" in settings panel. Analyzes page content (mock AI): Generates summary (2-3 sentences), key topics extracted (as tags), sentiment (Positive/Neutral/Negative), readability score. Display in settings panel.

27. SMART TAGS: After page has 100+ words, AI auto-suggests tags (mock: extract most common nouns). Show "Suggested tags" in settings with Accept/Reject per tag.

28. PAGE COMPARISON: From version history, select 2 versions. Opens side-by-side view. Differences highlighted: green for additions, red for deletions, yellow for changes.

29. CONTENT MAP: Click node opens that page. Lines drawn between pages that link to each other (from mentions/link-to-page blocks). Zoom in/out with +/- buttons. Drag canvas to pan. Node size reflects word count.

BOUNDARIES — CRITICAL:

- Do not add any database reads or writes
- All data in React state only (lost on refresh — expected)
- Do not change any visual design from Prompt 1
- Do not add any features not described above

PROMPT 4 OF 7

LOGIC: Database 6 Views + Collaboration + Search + Templates + AI Assistance

🔵 **REPLIT: AUTONOMOUS → ECONOMY | App Testing: OFF | Code Optimization: OFF**

> ✅ **FIX APPLIED:** Fix 1 — Tech stack corrected
> ✅ **FIX APPLIED:** Fix 3 — File paths corrected
> ✅ **FIX APPLIED:** Fix 15 — PDF export approach specified (window.print() or jspdf)

```
CONTEXT:
App: AI-driven productivity platform with auth, teams, tasks
Tech: React, TypeScript, Vite, Tailwind CSS, Shadcn/UI, PostgreSQL, Drizzle ORM, Express.js backend in server/routes.ts
Building: Advanced note-taking system with block editor, 35+ block types, databases, AI coaching, focus mode, voice notes, sketching, and 25 unique features beyond competitors
Feature: 6 database views, 20 property types, comments, reactions, sharing, version history, command palette, templates, import/export, AI text assistance
Current state: Prompt 3 done. All blocks and unique features work.
Layer: Layer 2+3
Files to change: client/src/components/notes/DatabaseView.tsx, client/src/components/notes/CollaborationPanel.tsx, client/src/components/notes/SearchCommandPalette.tsx, client/src/components/notes/PageSettingsPanel.tsx

1. DATABASE VIEWS (6 views sharing same data):
TABLE: Editable cells. Click cell to edit. Column resize. Sort by column click. Add row button. Add property (column) button. Property types: Text, Number, Select (dropdown), Multi-select, Status, Date, Person, Checkbox, URL, Email, Phone, Files, Formula, Relation, Rollup, Created time, Last edited, Created by, Last edited by.
BOARD: Cards in columns grouped by a Select/Status property. Drag cards between columns. Click card opens row detail.
CALENDAR: Monthly grid. Items placed on Date property. Click day to add. Drag to reschedule. Month navigation.
TIMELINE: Horizontal bars. Start/end from Date property. Drag to resize duration. Zoom levels.
GALLERY: Card grid with cover image from Files property. Title + 2-3 visible properties.
LIST: Simple row list with title + key properties inline. Compact.
Filter: Add filter rows (property + operator + value). AND/OR logic. Sort: Multi-column sort. Group: Group rows by any Select/Status property.

2. COMMENTS: Click comment icon on page header opens CollaborationPanel. Type + Enter posts comment with name + time. Reply to create thread. "Resolve" hides thread (show resolved toggle). Inline comments: Select text → toolbar "Comment" → adds highlight + comment.
```

3. BLOCK REACTIONS: Show reaction counts per block. Click to toggle your reaction.

4. SHARING: Share button opens modal. "Copy link" copies page URL. Permission dropdown: "Can edit" / "Can comment" / "Can view". Toggle "Public page" generates shareable link. "Invite" email input.

5. VERSION HISTORY: Triggered from settings "History". Shows list of auto-saved versions (every 10 minutes of editing). Click version shows page at that state. "Restore" replaces current with that version. Page comparison between versions.

6. COMMAND PALETTE: Ctrl+K. Search filters pages by title (case-insensitive). Actions execute. Recent tracks last opened. Quick Capture saves text to inbox. Results show matched pages with icon + title + snippet + path.

7. TEMPLATES: "Templates" button in new page flow. Gallery of templates: "Meeting Notes" / "Project Brief" / "Weekly Review" / "Decision Log" / "Research Notes" / "Daily Journal". Click creates page from template. "Save as template" from page menu.

8. IMPORT: Settings → Import. Accept .md files. Parse markdown to blocks. Drag-drop file onto page to import.

9. EXPORT: Settings → Export. Generate Markdown from blocks. Download as .md file. Export as PDF option: use the browser's window.print() method with a print-friendly stylesheet, or install jspdf for programmatic PDF generation.

10. AI ASSISTANCE (in BlockToolbar): Select text → sparkle button. Options: "Improve writing" / "Fix grammar" / "Make shorter" / "Make longer" / "Change tone (Professional/Casual/Friendly)" / "Summarize" / "Translate" / "Explain" / "Continue writing" / "Generate from prompt". Mock AI: return slightly modified text for now.

BOUNDARIES — CRITICAL:

- Do not add any database reads or writes
- All data in React state only (lost on refresh — expected)
- Do not change any visual design from Prompt 1
- Do not add any features not described above

PROMPT 5 OF 7

DB: Create ALL Tables + Connect Pages + Blocks + Sidebar

🔵 **REPLIT: AUTONOMOUS → ECONOMY | App Testing: OFF | Code Optimization: OFF**

> ✅ **FIX APPLIED:** Fix 1 — Tech stack corrected
> ✅ **FIX APPLIED:** Fix 2 — Existing 'notes' table conflict noted
> ✅ **FIX APPLIED:** Fix 3 — File paths corrected
> ✅ **FIX APPLIED:** Fix 4 — Drizzle pgTable() syntax (not Prisma)
> ✅ **FIX APPLIED:** Fix 5 — npx drizzle-kit push command added
> ✅ **FIX APPLIED:** Fix 6 — uuid, jsonb, date imports added
> ✅ **FIX APPLIED:** Fix 7 — UUID PKs use uuid().defaultRandom().primaryKey()
> ✅ **FIX APPLIED:** Fix 8 — User FK columns use text() to match Replit Auth
> ✅ **FIX APPLIED:** Fix 9 — workspace_id replaced with userId(text)
> ✅ **FIX APPLIED:** Fix 10 — API routes instruction added for server/routes.ts
> ✅ **FIX APPLIED:** Fix 12 — IStorage interface pattern added
> ✅ **FIX APPLIED:** Fix 13 — jsonb defaults use .$defaultFn()
> ✅ **FIX APPLIED:** Fix 19 — npx drizzle-kit push in boundaries

```
CONTEXT:
App: AI-driven productivity platform with auth, teams, tasks
Tech: React, TypeScript, Vite, Tailwind CSS, Shadcn/UI, PostgreSQL, Drizzle ORM,
Express.js backend in server/routes.ts
Building: Advanced note-taking system with block editor, 35+ block types, databases,
AI coaching, focus mode, voice notes, sketching, and 25 unique features beyond
competitors
Feature: Create every database table + connect pages, blocks, sidebar tree,
favorites, tags, recent to database
Current state: Prompts 1-4 done. Everything works with React state.
Layer: Layer 4 — Database

IMPORTANT: The app already has a 'notes' table in shared/schema.ts that handled
basic notes. The new 'pages' table replaces it. Do NOT delete the old notes table —
just create new ones alongside it.

Files to change: client/src/components/notes/Sidebar.tsx,
client/src/components/notes/PageView.tsx,
client/src/components/notes/BlockEditor.tsx,
client/src/components/notes/SlashCommandGrid.tsx, shared/schema.ts,
server/routes.ts, server/storage.ts

PREREQUISITE — SCHEMA IMPORTS:
First, add these to the existing import from drizzle-orm/pg-core in
shared/schema.ts:
```

uuid, jsonb, date

(Your file currently imports: pgTable, text, serial, integer, boolean, timestamp, varchar — add the missing ones.)

=== PART A: CREATE ALL 18 TABLES ===

Add all 18 tables to shared/schema.ts using Drizzle's pgTable() syntax.

Use uuid().defaultRandom().primaryKey() for all primary keys.

Use text() for all user ID foreign key columns (Replit Auth stores user IDs as text strings, not UUIDs).

Use .$defaultFn(() => ({})) for jsonb columns with default {}, and .$defaultFn(() => []) for default [].

Since there is no workspace system yet, replace all workspace_id references with userId as a text column referencing the authenticated user.

Do NOT connect to UI yet in Part A. Create tables first.

1. "pages": id(uuid PK), userId(text, FK to user), parent_page_id(uuid self-ref nullable), title(text default "Untitled"), icon(text nullable), cover_type(text nullable "gradient"/"image"), cover_value(text nullable), font(text default "default"), small_text(boolean false), full_width(boolean false), is_locked(boolean false), status(text default "draft"), is_favorite(boolean false), is_template(boolean false), is_public(boolean false), word_count(integer 0), sort_order(integer 0), last_visited_at(timestamp nullable), created_by(text, FK to user), created_at(timestamp), updated_at(timestamp), deleted_at(timestamp nullable)

2. "blocks": id(uuid PK), page_id(uuid FK pages), parent_block_id(uuid self-ref nullable for nested), type(text not null), content(text default ""), properties(jsonb .$defaultFn(() => ({}))), sort_order(integer 0), indent_level(integer 0), is_synced(boolean false), synced_source_id(uuid nullable), created_at(timestamp), updated_at(timestamp)

3. "block_reactions": id(uuid PK), block_id(uuid FK blocks), user_id(text FK), emoji(text), created_at(timestamp). UNIQUE(block_id, user_id, emoji)

4. "block_bookmarks": id(uuid PK), user_id(text FK), block_id(uuid FK blocks), page_id(uuid FK pages), label(text nullable), created_at(timestamp)

5. "page_tags": id(uuid PK), page_id(uuid FK pages), tag_name(text), is_ai_suggested(boolean false), created_at(timestamp)

6. "page_versions": id(uuid PK), page_id(uuid FK pages), version_number(integer), blocks_snapshot(jsonb), created_by(text FK), created_at(timestamp), description(text nullable)

7. "page_shares": id(uuid PK), page_id(uuid FK pages), shared_with_id(text FK nullable), shared_with_email(text nullable), permission(text default "view"), is_public_link(boolean false), created_by(text FK), created_at(timestamp)

8. "comments": id(uuid PK), page_id(uuid FK pages), block_id(uuid FK nullable), parent_comment_id(uuid self-ref nullable), author_id(text FK), content(text), is_resolved(boolean false), selection_text(text nullable), created_at(timestamp), updated_at(timestamp)

9. "databases_meta": id(uuid PK), block_id(uuid FK blocks), name(text), description(text nullable), created_at(timestamp), updated_at(timestamp)

10. "database_properties": id(uuid PK), database_id(uuid FK databases_meta), name(text), type(text), config(jsonb .$defaultFn(() => ({}))), sort_order(integer), is_visible(boolean true), created_at(timestamp)

11. "database_rows": id(uuid PK), database_id(uuid FK databases_meta), values(jsonb .$defaultFn(() => ({}))), created_by(text FK), created_at(timestamp), updated_at(timestamp), deleted_at(timestamp nullable)

12. "database_views": id(uuid PK), database_id(uuid FK databases_meta), name(text), type(text "table"/"board"/"calendar"/"timeline"/"gallery"/"list"), config(jsonb), is_default(boolean false), sort_order(integer), created_at(timestamp)

13. "snippets": id(uuid PK), user_id(text FK), name(text), blocks_data(jsonb), created_at(timestamp), updated_at(timestamp)

14. "captures": id(uuid PK), user_id(text FK), content(text), source(text nullable), is_processed(boolean false), created_at(timestamp)

15. "journal_entries": id(uuid PK), user_id(text FK), date(date), page_id(uuid FK pages), mood(text nullable), streak_count(integer default 0), created_at(timestamp)

16. "page_insights": id(uuid PK), page_id(uuid FK pages), summary(text), topics(text[]), sentiment(text), readability_score(integer), generated_at(timestamp)

17. "page_templates": id(uuid PK), userId(text FK), name(text), description(text), icon(text), blocks_data(jsonb), category(text), usage_count(integer 0), created_by(text FK), created_at(timestamp)

18. "recent_pages": id(uuid PK), user_id(text FK), page_id(uuid FK pages), visited_at(timestamp). UNIQUE(user_id, page_id)

After adding all tables to shared/schema.ts, run: npx drizzle-kit push

=== PART B: CONNECT PAGES + BLOCKS + SIDEBAR ===

IMPORTANT: For all database operations, add methods to the IStorage interface in server/storage.ts and implement them there, then call storage.* methods from routes — this is the existing pattern in the app. Add all necessary API routes to server/routes.ts following the same patterns as existing routes, using isAuthenticated middleware and req.user.claims.sub for user ID.

After tables are created, connect these features to the database:

1. PAGES: Create → Insert into pages with userId from req.user.claims.sub. Load page tree: Load all pages for user, build tree from parent_page_id. Update

title/icon/cover/font/width/small_text/status/lock on change. Delete → Set deleted_at (soft delete). Restore from trash → Clear deleted_at.

2. BLOCKS: On page open → Load blocks for page_id ordered by sort_order. Create block → Insert. Update content/type/properties on change (debounced 1 second). Delete block → Delete row. Reorder (drag-drop) → Update sort_order for affected blocks. Indent/outdent → Update indent_level.

3. SIDEBAR TREE: Load from pages table. Drag reorder → Update sort_order + parent_page_id. Expand/collapse state stored in localStorage.

4. FAVORITES: Star → Update is_favorite on page. Load favorites tab: pages where is_favorite=true.

5. TAGS: Add tag → Insert page_tags. Remove → Delete. Load Tags tab: Get distinct tag_name from page_tags, count pages per tag. Click tag → Load pages with that tag.

6. RECENT: On page open → Upsert recent_pages (user+page, update visited_at). Load Recent tab: Order by visited_at desc limit 10.

7. QUICK CAPTURE: Save → Insert into captures. Load inbox → captures where is_processed=false. "Move to page" → Create block from capture, set is_processed=true.

8. VERSION AUTO-SAVE: Every 10 minutes of editing → Snapshot all blocks as JSON, insert page_versions. Increment version_number. Keep last 50 versions.

BOUNDARIES — CRITICAL:

- Do not change any UI design or layout
- Do not change any interaction behavior
- Only add database operations (schema, storage methods, API routes)
- After adding all tables, run: npx drizzle-kit push
- Do not add any features not described above

PROMPT 6 OF 7

DB: Connect Database + Collaboration + Search + Unique Features + AI + Templates

🔵 **REPLIT: AUTONOMOUS → ECONOMY | App Testing: OFF | Code Optimization: OFF**

> ✅ **FIX APPLIED:** Fix 1 — Tech stack corrected
> ✅ **FIX APPLIED:** Fix 3 — File paths corrected
> ✅ **FIX APPLIED:** Fix 9 — userId instead of workspace_id
> ✅ **FIX APPLIED:** Fix 10 — API routes instruction added
> ✅ **FIX APPLIED:** Fix 12 — IStorage interface pattern added
> ✅ **FIX APPLIED:** Fix 18 — Content recycler handles NULL last_visited_at

```
CONTEXT:
App: AI-driven productivity platform with auth, teams, tasks
Tech: React, TypeScript, Vite, Tailwind CSS, Shadcn/UI, PostgreSQL, Drizzle ORM,
Express.js backend in server/routes.ts
Building: Advanced note-taking system with block editor, 35+ block types, databases,
AI coaching, focus mode, voice notes, sketching, and 25 unique features beyond
competitors
Feature: Connect databases, comments, reactions, sharing, search, snippets, journal,
insights, templates, AI, content recycler, synced blocks, import/export — ALL
remaining features
Current state: Prompt 5 done. Pages + blocks persist. All tables exist.
Layer: Layer 4
Files to change: client/src/components/notes/DatabaseView.tsx,
client/src/components/notes/CollaborationPanel.tsx,
client/src/components/notes/SearchCommandPalette.tsx,
client/src/components/notes/PageSettingsPanel.tsx,
client/src/components/notes/FocusMode.tsx, client/src/components/notes/Sidebar.tsx,
client/src/components/notes/BlockEditor.tsx, server/routes.ts, server/storage.ts


IMPORTANT: For all database operations, add methods to the IStorage interface in
server/storage.ts and implement them there, then call storage.* methods from routes.
Add all necessary API routes to server/routes.ts following the same patterns as
existing routes, using isAuthenticated middleware and req.user.claims.sub for user
ID.


=== PART A: DATABASE + COLLABORATION + SEARCH ===
1. DATABASES: When database block created → Insert databases_meta + default
properties (Name text, Status select). Add property → Insert database_properties.
Add row → Insert database_rows. Edit cell → Update database_rows.values jsonb.
Delete row → soft delete. Save view config → Insert/update database_views.
2. COMMENTS: Post → Insert comments. Load → by page_id ordered created_at. Reply →
Insert with parent_comment_id. Resolve → Update is_resolved. Inline comment → Set
block_id + selection_text.
```

3. BLOCK REACTIONS: Add → Insert block_reactions. Remove → Delete. Load → by block_id grouped by emoji with count.

4. BLOCK BOOKMARKS: Save → Insert block_bookmarks. Load user's bookmarks → Join with blocks + pages for display.

5. SHARING: Share → Insert page_shares. Update permission. Toggle public → Update is_public_link. Load shares for page.

6. SEARCH: Full-text search across pages.title + blocks.content. Return matched pages with snippet of matching text. Filter by tag, date range, creator. Order by relevance (title match > content match).

7. VERSION HISTORY: Load versions for page ordered desc. Click version → Render blocks_snapshot read-only. Restore → Replace current blocks with snapshot, create new version.

=== PART B: UNIQUE FEATURES + AI + TEMPLATES ===

8. SNIPPETS: Save snippet → Insert snippets with blocks_data jsonb. Load library → by user. Insert snippet → Create blocks from blocks_data at cursor position. Delete snippet → Delete row.

9. JOURNAL: "Today's Journal" → Check if journal_entries exists for today. If yes: open linked page. If no: Create page from journal template, insert journal_entries row. Streak: Count consecutive days with entries. Display in sidebar.

10. INSIGHTS: "Generate Insights" → Collect all block content for page. Send to AI (mock: generate summary from first 200 chars, extract nouns as topics, random sentiment). Insert/update page_insights. Display in settings.

11. SMART TAGS: After page save with 100+ words → Extract common words (mock AI). Insert page_tags with is_ai_suggested=true. Show in settings for accept/reject.

12. TEMPLATES: Save page as template → Snapshot blocks, insert page_templates. Load template gallery → by user + built-in. Create from template → Create page + blocks from blocks_data.

13. CONTENT RECYCLER: Query pages WHERE (last_visited_at < 30 days ago OR last_visited_at IS NULL) AND deleted_at IS NULL AND word_count > 50. Show top 3 in sidebar "Rediscover" section. Dismiss → Update last_visited_at to now.

14. SYNCED BLOCKS: On sync create → Set is_synced=true, synced_source_id to original. On source edit → Update all blocks where synced_source_id matches. Unsync → Set is_synced=false, clear synced_source_id.

15. IMPORT: Parse .md file to block array. Insert page + blocks.

16. EXPORT: Load page blocks, convert to Markdown string, trigger download.

17. AI TOOLBAR: Select text → Sparkle → Action. Mock AI processing: "Improve" returns slightly reworded text. "Summarize" returns first sentence. "Continue" appends lorem text. Save changed text to block content.

18. FOCUS MODE STATS: Track focus sessions: start time, end time, words written. Store in localStorage (or user preferences). Display streak in focus mode UI.

BOUNDARIES — CRITICAL:

- Do not change any UI design or layout
- Do not change any interaction behavior
- Only add database operations, storage methods, and API routes
- Do not add any features not described above

PROMPT 7 OF 7

DB: Verification + Seed Content (FINAL)

🔵 **REPLIT: AUTONOMOUS → ECONOMY | App Testing: OFF | Code Optimization: OFF**

> ✅ **FIX APPLIED:** Fix 1 — Tech stack corrected

```
CONTEXT:
App: AI-driven productivity platform with auth, teams, tasks
Tech: React, TypeScript, Vite, Tailwind CSS, Shadcn/UI, PostgreSQL, Drizzle ORM,
Express.js backend in server/routes.ts
Building: Advanced note-taking system with block editor, 35+ block types, databases,
AI coaching, focus mode, voice notes, sketching, and 25 unique features beyond
competitors
Feature: Verify all connections + seed sample content for testing
Current state: Prompt 6 done. All features connected.
Layer: Layer 4 — Final prompt

TASK:
Verify every save/load works. Seed rich sample content so the platform looks
populated.

1. VERIFY ALL CONNECTIONS (test each one, fix if broken):
- Create page → appears in sidebar after refresh
- Edit title/icon/cover → persists
- Add blocks of every type → all persist with correct content/properties
- Drag reorder blocks → persists
- Nest page under another → tree updates correctly
- Star/unstar → favorites tab updates
- Add/remove tags → tags tab updates
- Create database → properties + rows persist
- Switch database views → configs persist
- Post comment → persists, reply threads work
- Add block reaction → persists with count
- Bookmark block → appears in bookmarks collection
- Share page → permission saved
- Version history shows auto-saved versions
- Save snippet → insert snippet works
- Journal creates daily page with streak
- Search returns matching pages
- Template gallery loads, create from template works
- Export downloads .md, Import creates page from .md
- Quick capture saves to inbox
- Content recycler surfaces old pages (including pages with NULL last_visited_at)
```

2. SEED SAMPLE CONTENT (create all of this):

- Page "Getting Started" with 8 varied blocks (text, heading, bullet, callout, code, image placeholder, divider, toggle with children)
- Page "Project Notes" with 2 sub-pages "Sprint 1" (with task database inline) and "Sprint 2"
- Page "Meeting Notes" with 3 sub-pages each using meeting template
- Page "Ideas" with flashcard block + poll block + progress tracker
- Page "Research" with reading list block + voice note placeholder + timeline block
- 1 inline database "Task Tracker" with 3 properties (Name, Status, Priority) + 5 sample rows + Table + Board views
- 5 sample comments across different pages
- 3 block reactions
- 3 snippets ("Meeting Template", "Decision Log", "Quick Notes")
- 6 page templates (Meeting Notes, Project Brief, Weekly Review, Decision Log, Research Notes, Daily Journal)
- 5 tags (Work, Personal, Ideas, Research, Team)
- 3 favorites
- 2 quick captures in inbox
- 1 journal entry for today

BOUNDARIES — CRITICAL:

- Do not change any UI design or layout
- Do not change any interaction behavior
- Only add database operations and seed data
- Do not add any features not described above